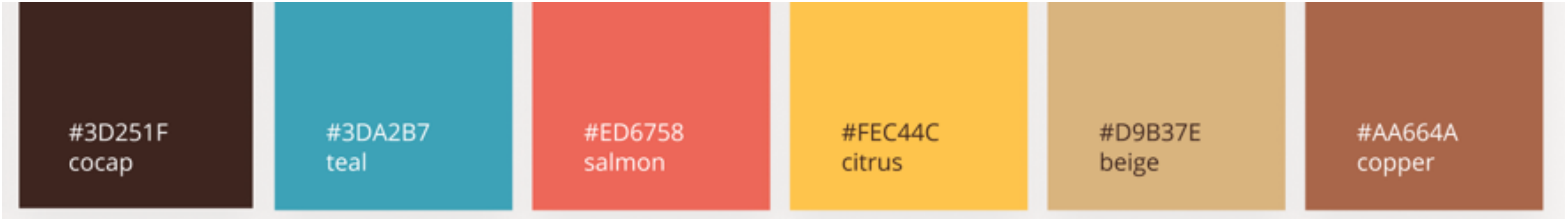
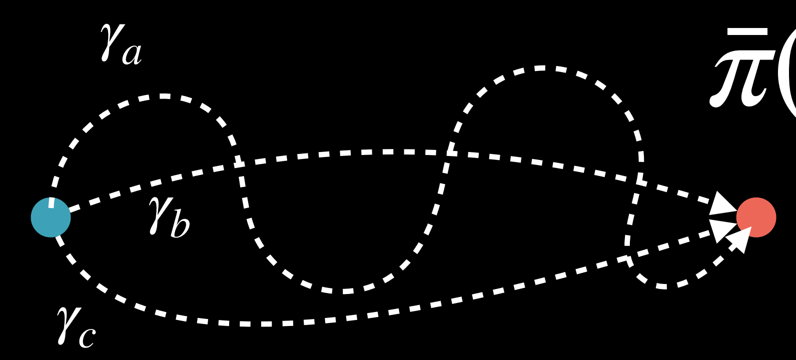

#3D251F
cocap
#3DA2B7
teal
#ED6758
salmon
#FEC44C
citrus
#D9B37E
beige
#AA664A
copper

Dynamic Formulation on Path Space



- Consider directly the space of paths over $\mathscr{X}$, i.e., $\bar{\mathscr{X}} = C([0,1], \mathscr{X})$

- The dynamic of 'particles' between $\alpha = \rho_0, \beta = \rho_1$ is given by distributions $\bar{\pi} \in \mathscr{P}(\bar{\mathscr{X}})$

- Valid distributions must have the right marginals, i.e., $(P_0)_\# \bar{\pi} = \rho_0, \quad (P_1)_\# \bar{\pi} = \rho_1,$ where $P_t(\gamma) = \gamma(t)$. Let $\overline{\Pi}(\alpha, \beta)$ be the set of all such $\bar{\pi}$.

- With this, can reformulate dynamic OT over the space of paths:

$$\mathrm{W}_2^2(\rho_0, \rho_1) = \min_{\bar{\pi} \in \overline{\Pi}} \int_{\bar{X}} L(\gamma)^2 \, d\bar{\pi}(\gamma)$$

where $L(\gamma) = \int_0^1 |\dot{\gamma}(s)|^2 \mathrm{d}s =$ length of path γ

Dynamic Formulation on Path Space